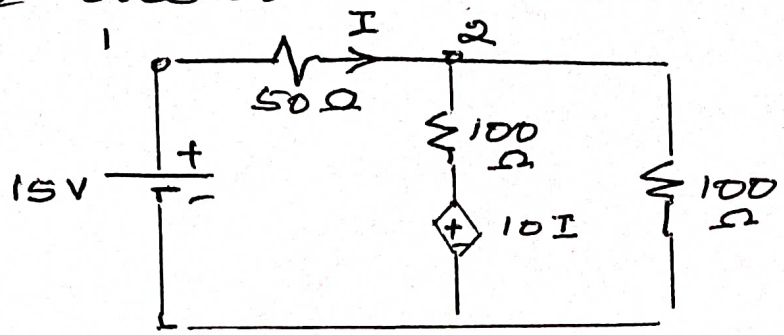
NETWORK ANALYSIS
ECE 215J
MID TERM EXAMINATION
OCT 2020.
III ECE
SCHEME OF VALUATION

Q 1. Find the voltage at node 2 of the network shown. **2M**



Soln. KCL at node 2

$$\frac{V_2 - 15}{50} + \frac{V_2 - 10I}{100} + \frac{V_2}{100} = 0. \quad \text{--(1)}$$

where $I = \frac{15 - V_2}{50} \Rightarrow 10I = \left(3 - \frac{V_2}{5}\right)$

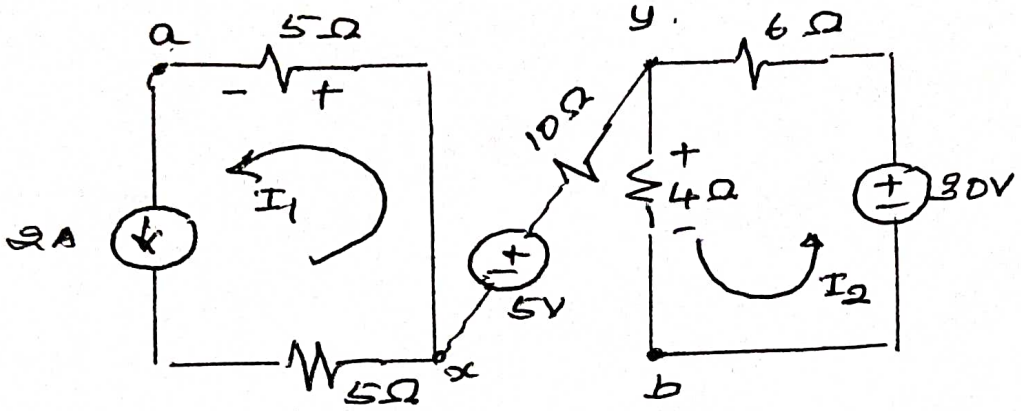
Eq (1):

$$\frac{V_2 - 15}{50} + \frac{V_2 - 3 + \frac{V_2}{5}}{100} + \frac{V_2}{100} = 0.$$

$$V_2\left[\frac{1}{50} + \frac{1}{100} + \frac{1}{500} + \frac{1}{100}\right] = \frac{15}{50} + \frac{3}{100}.$$

$$\therefore V_2 = 7.86\ V$$

Q2 Find the voltage Vab in the network shown in Fig. **2M**



Soln.

$I_1 = 2A$

$I_2 = \frac{30}{10} = 3A$

$V_{ab} = -I_1 \times 5 - 5 + I_2 \times 4$

$= -2 \times 5 - 5 + 3 \times 4$

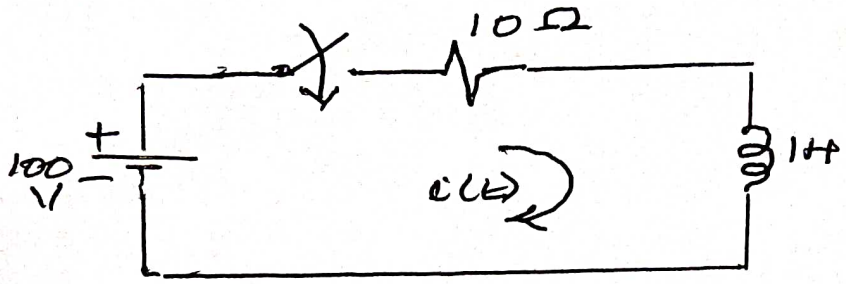
$= -10 - 5 + 12$

$V_{ab} = -3V$

Q 3 In the circuit shown in Fig the switch is closed at $t=0$. Find $i(0+)$, $\frac{di}{dt}(0+)$ and $i(\infty)$ 2 m

$\frac{1}{2} \times 3 = 1\frac{1}{2}$ m

$\frac{1}{2}$ m: steps



Soln.

$i(0-) = 0$.

$\therefore i(0+) = i(0-) = 0$ A

For $t > 0$.

$$10i + 1 \cdot \frac{di}{dt} = 100$$

At $t = 0+$,

$$10\, i(0+) + \frac{di}{dt}(0+) = 100$$

$$\therefore \frac{di}{dt}(0+) = 100 \text{ A/s}$$

At $t = \infty$, inductor acts as a S.C $\therefore i(\infty) = \frac{100}{10} = 10$ A

.. 4 |=

Q4p. Calculate the power supplied by the source in the network in Fig. 3 ans.

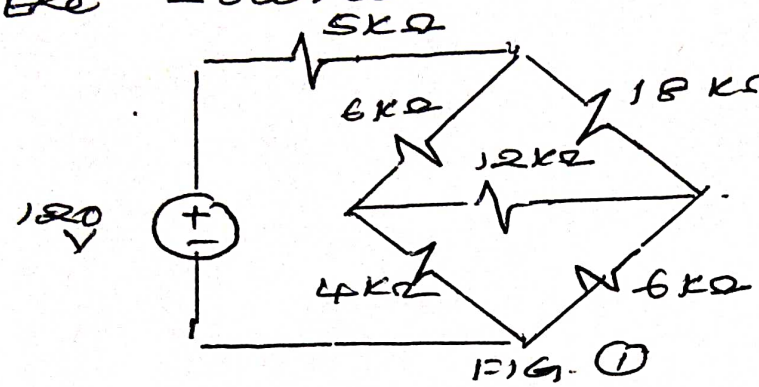


FIG. (1)

R_{eq} = 2 ans.
Power = 1 ans.

Solⁿ.

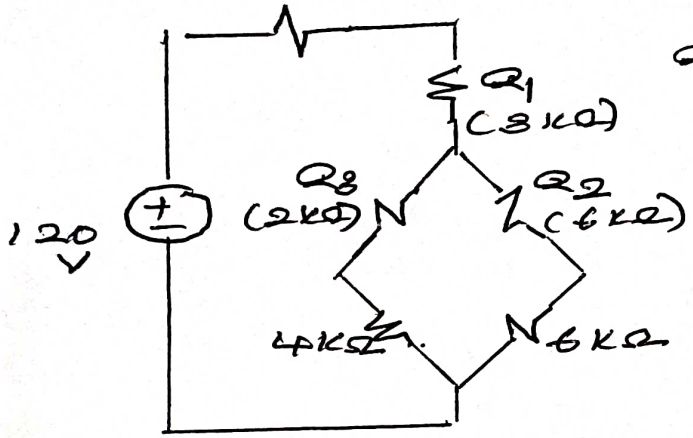


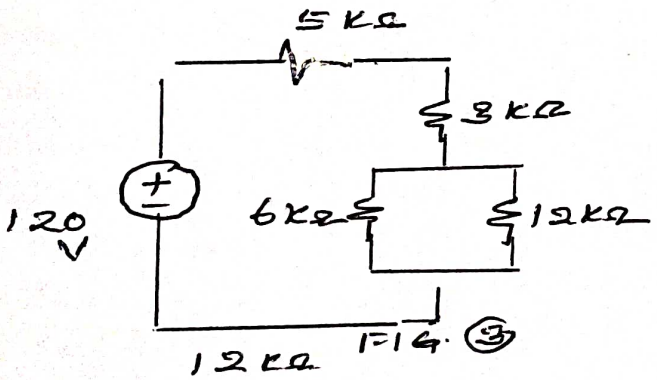
FIG. (2).

STEP (1) Delta to star:

$R_1 = \frac{6 \times 18}{36} = 3\,K\Omega$

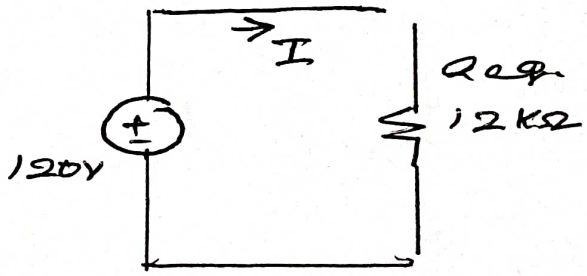
$R_2 = \frac{18 \times 12}{36} = 6K\Omega$

$R_3 = \frac{12 \times 6}{36} = 2K\Omega$



12 KΩ ∥ 6KΩ = 4KΩ

FIG. (3)



$P = I^2 . R$

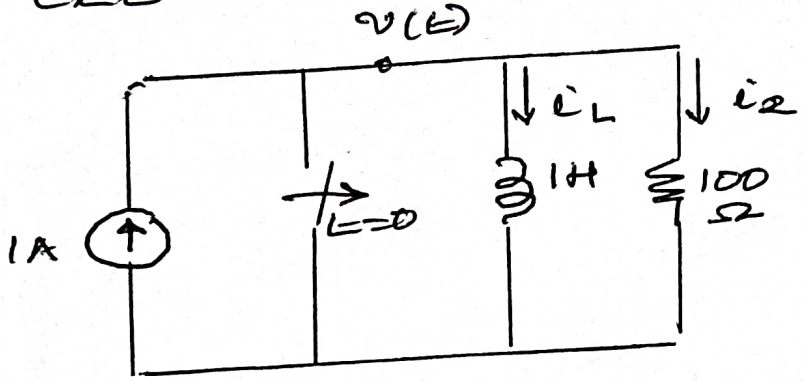
$= \left[\frac{120}{12 \times 10^3}\right]^2 \times 12 \times 10^3$

$= \left[10 \times 10^{-3}\right]^2 \times 12 \times 10^3$

$= 120 \text{ mW}$

$= 1.2 \text{ W}$.

Q 5 Find i_2 at $t = 20$ ms in the ckt shown in Fig. [3M]



$i_{2n} = 1M$

$i_{2f} = 1M$

$i_2(20ms) = 1M.$

Solⁿ: For $t < 0$, $i_L(0^-) = 0 = i_L(0^+)$.

$\tau = L/R = 1/100 = 0.01$ s.

$i_2(t) = i_{2n} + i_{2f}$ ∴ i_{2n} = natural response

i_{2f} = Forced response

$= i_2(\infty) = 0$ A

$\therefore i_2(t) = i_2(\infty) - t/\tau$

$= Ke^{-t/\tau} = Ke^{-t/0.01}$.

$i_2(0^+) = 1 = K.e^{-t/0.01}.A$

$\therefore i_2(t) = e^{-t/0.01}$

At $t = 20$ ms.

$i_2(t) = e^{-20\times10^{-3}/0.01}$

$= e^{-2}$

$= 0.135$ A

Q6. Find R_L for max. power transfer and the max. power that can be transfered to the load in the circuit of Fig. [4M]

V_{TH} : 1½ m
R_{TH} : 1½ m
$P_{L\,max}$: 1m.

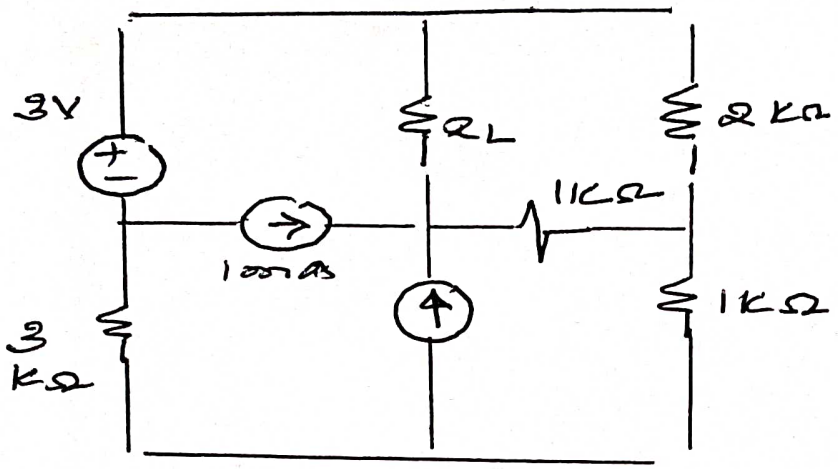


FIG. (1).

Soln.

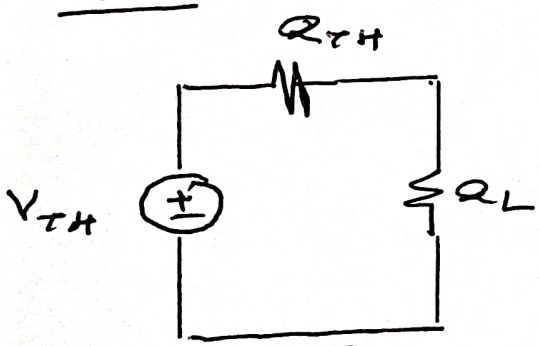
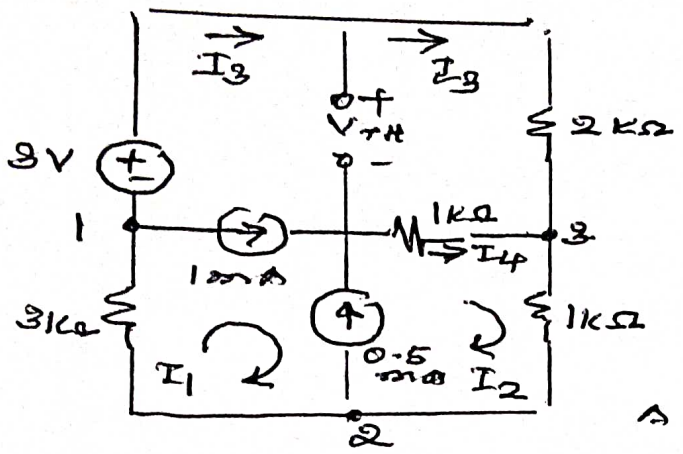


FIG. (2).
THEVENIN'S EQ. CKT.

(a) TO FIND V_{TH}.
(b) TO FIND R_{TH}.
WHEN $R_L = R_{TH}$ MAX. POWER GETS DELIVERED TO THE LOAD.

(a) TO FIND V_{TH} :



$V_{TH} = I_3 \times 2 - I_4 \times 1$

$V_{TH} = 2I_3 - I_4 \times 1$ — (1)

Loop $I_4 = 1 + 0.5 = 1.5$ mA

$V_{TH} = 2I_3 - 1.5$ — (2)

At node (1):

$I_1 = I_3 + 1$

$\therefore I_3 = I_1 - 1$ — (3)

At node (2):

$I_2 = 0.5 + I_1$

$I_2 - I_1 = 0.5$ — (4)

Outer loop:

$2I_3 + I_2 + 3I_1 = 3$ — (5)

At node (3) : $I_3 + I_4 = I_2$

$\therefore I_3 = I_2 - 1.5$

$\therefore$ Eq (5):

$2(I_2 - 1.5) + I_2 + 3I_1 = 3$

$3I_2 + 3I_1 = 6$

$I_2 + I_1 = 2$ — (6)

Eq (4) $I_2 - I_1 = 0.5$

$\therefore I_2 = 1.25$ mA

$I_1 = 0.75$ mA

Eq (3) $I_3 = I_1 - 1 = -0.25$ mA

Eq (2) $V_{TH} = 2I_3 - 1.5$

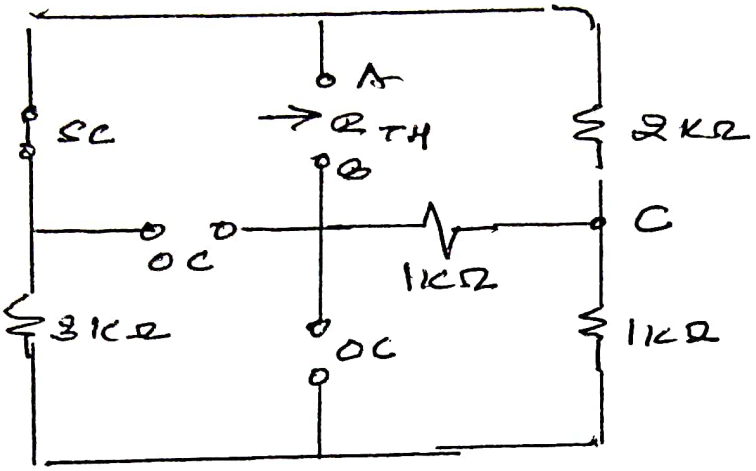
$= -2$V

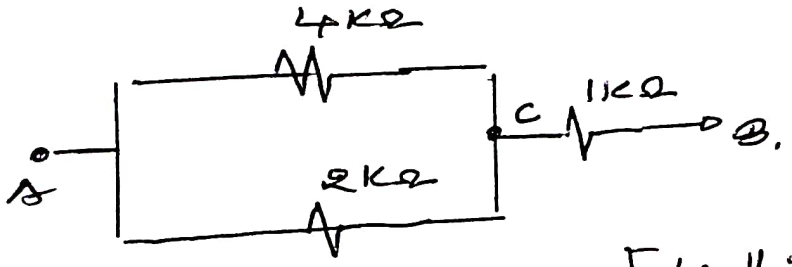
(b) TO FIND R_{TH} :

Remove the load

S/C the voltage source

O/C the current source



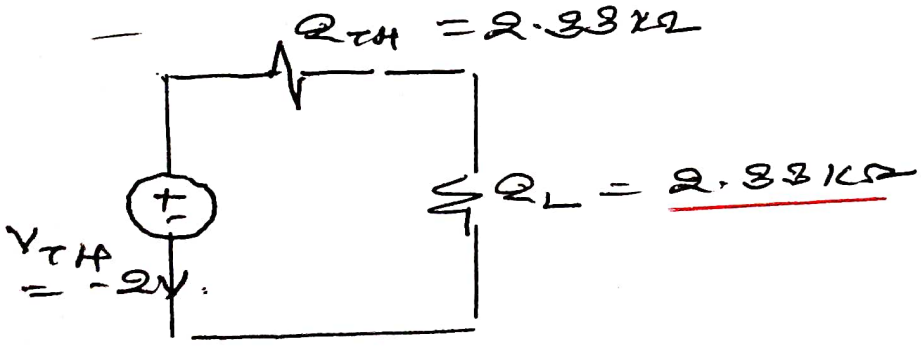


$\therefore R_{TH} = R_{AB} = [4 \parallel 2] + 1$

$= \frac{4 \times 2}{6} + 1$

$= 4/3 + 1$

$R_{TH} = \underline{2.33\,K\Omega}$



When $R_L = R_{TH} = 2.33\,K\Omega$ max. power gets delivered to the load.

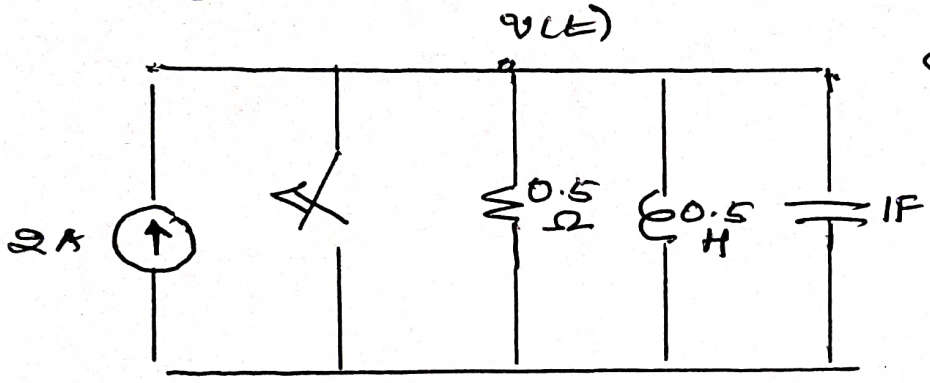
$P_{L\,(max)} = I_L^2 \cdot R_L$

$= \frac{V_{TH}^2}{4R_L^2} \cdot R_L$

$= \frac{V_{TH}^2}{4R_L} = \frac{4}{4 \times 2.33}$

$= \underline{0.429\,mW}$.

Q 7. In the circuit shown in Fig the switch is opened at $t=0$. Obtain the expression for $v(t)$ for $t>0$. [4M]



General soln : 2 on
Particular soln : 2 on.

Soln.

$$i_L(0^-) = i_L(0^+) = 0 \; ; \quad v_c(0^-) = v_c(0^+) = 0V$$

For $t>0$, KCL

$$\frac{v(t)}{0.5} + \frac{1}{0.5}\int_0^t v(t)\,dt + 1\cdot\frac{dv}{dt} = 2 \qquad -(1)$$

Diff.

$$2\frac{dv}{dt} + 2v + \frac{d^2v}{dt^2} = 0$$

$$\frac{d^2v}{dt^2} + 2\frac{dv}{dt} + 2v = 0$$

$$s^2 + 2s + 2 = 0.$$

$$s_1, s_2 = -1 \pm j1$$

$$v(t) = e^{-t}(K_1\cos t + K_2 \sin t) \qquad -(2).$$

$$\frac{dv}{dt} = -e^{-t}(K_1\cos t + K_2\sin t) + e^{-t}[-K_1\sin t + K_2\cos t]$$

$$\frac{dv}{dt} = e^{-t}[-K_1(\cos t + \sin t) + K_2(\cos t - \sin t)] \qquad -(3)$$

At $t = 0$,

$$v(0) = 0$$

From (2): $v(0) = 0 = K_1$

$$\frac{dv}{dt}(0+) = -K_1 + K_2 \quad [\text{Eq. (3)}]$$

From (1) At $t = 0+$,

$$0 + 0 + \frac{dv}{dt}(0+) = 2$$

$$\therefore \frac{dv}{dt}(0+) = 2 \text{ V/s}$$

$$\therefore K_1 = 0.$$

$$2 = -K_1 + K_2$$

$$\therefore K_2 = 2$$

$\therefore$ From (2):

$$v(t) = 2e^{-t}\sin t \quad ; t > 0.$$

Q7) Step1:-

METHOD (2)

$$\omega_0 = \frac{1}{\sqrt{LC}} = \sqrt{2}$$

$$\alpha = \frac{1}{2RC} = 1$$

$\omega_0 > \alpha$ underdamped. → 1

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 1$$

$$\text{Response} = v_C(t) = e^{-\alpha t}(B_1 \cos\omega_d t + B_2 \sin\omega_d t)$$

Step2 :-

$$i_L(0^-) = i_L(0^+) = 0$$

$$v_C(0^-) = v_C(0^+) = 0.$$ → 1

Step3 :-

$$v_C(t) = e^{-t}(B_1 \cos t + B_2 \sin t)$$

$$v_C(0) = B_1 = 0$$

$$\left.\frac{dv_C}{dt}\right|_{t=0^+} = B_2\omega_d - B_1\alpha = \frac{i_C(0^+)}{C}$$

$$B_2 = 2$$

$i_C(0^+) = 2A$

} 4 2

Step4 :-

$$v_C(t) = e^{-t}(2\sin t)$$

$$\underline{v_C(t) = 2e^{-t}\sin t}$$

— END —